

SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

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Abstract: We study a varying coefficient model in which the effect of each linear covariate is additive in several modulating covariates, and each additive component is expanded in a periodized wavelet basis. The resulting design is an approximation space of products between a linear covariate and a wavelet evaluated at a modulator, and the components are estimated by a single ~~lasso~~ block lasso problem in which the levels of the coefficients are left unpenalized and the wavelet coefficients of each resolution level are grouped in chunks of order $\log n$. Three results are established. The population Gram matrix of the product design has its smallest eigenvalue bounded away from zero under a conditional second moment condition on the linear covariates and a bounded joint density for the modulators, with no independence between the two groups of covariates; the estimator satisfies a nonasymptotic oracle inequality that needs no cone condition, because the unpenalized levels are profiled out, and that holds for any weights of bounded ratio; and, at a resolution level chosen inside an explicit window, the prediction error and the componentwise error attain the rate of nonlinear wavelet approximation ~~up to a logarithmic factor~~, with no logarithmic factor when the

Besov integrability index is at least two, which is minimax optimal, under the same Besov condition that governs linear approximation. The last result rests on the observation that the Besov condition already implies compressibility of the wavelet coefficients a bound on the ideal risk over chunks that the Besov condition already implies, so that adaptivity costs no additional assumption. A threshold on the norms of the fitted components recovers which modulators act on which coefficients, and the thresholded estimator keeps the rate of the fit. The estimator is a single convex program and is fit by coordinate descent on a sparse design matrix standard group lasso software. An implementation in R and the code that reproduces every numerical result of the paper are made available with it.

Key words and phrases: Additive coefficient model, Besov space, effective dimension, group lasso, lasso, oracle inequality, thresholding, varying coefficient model, wavelets.

1. Introduction

Applied problems often present an effect that is not constant across the sample. The return to schooling changes with age and with experience, the effect of temperature on electricity demand changes with the hour of the day and with humidity, and the effect of a treatment changes with the age and the body mass index of the patient. The varying coefficient model of Hastie and Tibshirani (1993) and Cai, Fan, and Yao (2000) accommodates this by letting the coefficient of each covariate be a function of a modulating variable; see Fan and Zhang (2008) for a review. When more than one

variable modulates the effect, a fully nonparametric coefficient runs into the curse of dimensionality, and the standard remedy is to make each coefficient additive in the modulators, which is the additive coefficient model of Xue and Yang (2006), studied further by Xue and Liang (2010) and Liu and Yang (2010).

The estimators available for this model are built from polynomial splines and from kernels, and are governed by a global smoothness index, so that a component with a peak, a jump or a change of regime is either over-smoothed or paid for everywhere in the domain. Wavelet bases combined with an ℓ_1 penalty (Tibshirani, 1996) are the standard answer to this difficulty, because a function whose smoothness is inhomogeneous is still economically represented by a few large coefficients (Donoho and Johnstone, 1994, 1998; Antoniadis and Fan, 2001). They have been used for varying coefficients without an additive structure and without penalization (Zhou and You, 2004; Montoril, Morettin, and Chiann, 2018), and for additive models without a multiplying covariate (Sardy and Tseng, 2004; Zhang and Wong, 2003; Haris et al., 2018; Amato et al., 2022; Sardy and Ma, 2024); penalized estimation in the additive coefficient model itself has been carried out with grouped penalties on spline bases (Wei et al., 2011; Antoniadis et al., 2014; Ma et al., 2015), again under global smoothness.

Klopp and Pensky (2015) study the varying coefficient model in which a single index modulates every coefficient. They expand each coefficient function in an orthonormal basis, estimate the resulting array by a block lasso, and obtain a nonasymptotic oracle inequality, a Besov rate that adapts to inhomogeneous smoothness, and a matching minimax lower bound, under the assumption that the covariates are independent of the index. The present paper carries that ~~programme~~ [program](#) to the situation an applied problem usually presents, in which several variables modulate a coefficient at once. We let each coefficient be additive in the modulators, which keeps the estimator free of the curse of dimensionality in their number, we allow the covariates to depend on the modulators, and we ~~penalize coefficient~~ [by coefficient](#) rather than in blocks, leaving the levels of the coefficients unpenalized. Each of these costs something: [keep the block lasso but leave the levels of the coefficients unpenalized and group the wavelet coefficients in chunks of order \$\log n\$ within a resolution level, with weights whose ratio stays bounded, so that the standardized weights of group lasso software are covered](#). Additivity produces cross terms between distinct modulators, so the population Gram matrix no longer factorizes as a Kronecker product and has to be bounded directly; dependence between the covariates and the modulators replaces a marginal second moment by a conditional one; and

the unpenalized levels leave a term in the risk that full block penalization does not incur. Our main result, Theorem 1, states that the estimator attains the rate of nonlinear wavelet approximation up to a logarithmic factor, under the same Besov assumption that governs linear approximation, and it reduces to the rate of Klopp and Pensky when a single modulator is present. Klopp and Pensky, with no logarithmic factor when the Besov integrability index is at least two, so that with a single modulator independent of the covariates it is minimax optimal there where it is minimax optimal. It holds under weaker conditions than theirs: the number of covariates is fixed, a regime their high dimensional condition excludes, and the effective smoothness of the components need only exceed $s/(2s + 1)$, which is below one half, where theirs must exceed one half. Components with effective smoothness one half, such as the inhomogeneous functions of Section 5, are covered here and not there. A lower bound for several modulators remains open. The lower bound that makes it the rate optimal is proved here for any number of modulators and for covariates that depend on them; Klopp and Pensky have it with a single modulator independent of the covariates.

We call the resulting procedure WAFC, for wavelet additive functional coefficients. Its contributions are of three four three kinds. The first is the theory of the additive product design: a design condition for the Gram

matrix of the products between a linear covariate and a wavelet of a modulator, which carries a cross term between distinct modulators and does not assume independence (Proposition 2); an oracle inequality that needs no cone condition, because the unpenalized levels are profiled out, and that holds for any weights of bounded ratio, including those of the software that computes the estimator (Theorem 2); and rates obtained by compressibility that follow from the Besov assumption alone, through a bound on the ideal risk over chunks (Theorem 1), together with a minimax lower bound, for any number of modulators, that makes these rates optimal when $\pi \geq 2$ (Theorem S6.1 in the Supplementary Material). The second is the reading that compressibility costs no assumption: the Besov condition on the components already implies that their wavelet coefficients lie in a weak ℓ_τ ball with $\tau = (s + 1/2)^{-1}$ (Lemma 3, a form with explicit constants of the classical embedding in Johnstone, 2019, Proposition 9.15), and it is this that separates the rate of the estimator from the rate of linear approximation when the Besov integrability index is smaller than two. The second is the recovery of the structure of the model, which modulators act on which coefficients, by a threshold on the norms of the fitted components (Corollary 2): removing the null blocks needs no separation condition, and the thresholded estimator keeps the rate of the fit (Corollary 3), so that one

estimator serves both for structure and for prediction. The third is computational: the estimator is one convex program ~~on a sparse design matrix~~ solved by standard group lasso software, and the resolution level and the penalty are chosen jointly by cross-validation, and the threshold by the one standard error rule on the same folds. ~~The fourth is the recovery of the structure of the model, which modulators act on which coefficients, by a threshold on the norms of the fitted components (Corollary 2): it needs no design condition beyond the one above and no irrepresentability condition, and removing the null blocks needs no separation condition at all.~~

The rest of the paper is organized as follows. Section 2 introduces the model, its identification, the wavelet approximation space and the estimator. Section 3 states the assumptions and the theory, opening with the main rate and then giving the three results it rests on, then the rates, including one with no design condition, and closes with the recovery of the block structure by thresholding and the risk of the thresholded estimator. Section 4 describes the computation, the scale of the penalty and the tuning rules, the threshold among them. Proofs are collected in the online Supplementary Material, which also gives the theory of the coordinatewise lasso, offered by the software as an option.

Throughout, $\|a\|_\pi$ is the ℓ_π norm of a vector, $\|h\|_n^2 = n^{-1} \sum_{i \leq n} h_i^2$ is

the empirical norm, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are the extreme eigenvalues of a symmetric matrix, $\|\cdot\|_{\text{op}}$ is the operator norm, and $a_n \asymp b_n$ means that a_n/b_n is bounded away from zero and infinity. For a support S , δ_S is the vector that agrees with δ on S and vanishes elsewhere.

2. Model and Wavelet [Approximation Space](#)

2.1 The additive functional coefficient model

Let Y be a scalar response, $\mathbf{X} = (X_1, \dots, X_p)^\top$ a vector of linear covariates and $\mathbf{U} = (U_1, \dots, U_q)^\top$ a vector of modulating covariates, taking values in $[0, 1]^q$ after a monotone transformation of each coordinate. The model is

$$Y = \sum_{\ell=1}^p \beta_\ell(\mathbf{U}) X_\ell + \varepsilon, \quad \mathbb{E}(\varepsilon \mid \mathbf{X}, \mathbf{U}) = 0, \quad (2.1)$$

with each functional coefficient additive in the modulators,

$$\beta_\ell(u) = c_\ell + \sum_{m=1}^q g_{\ell m}(u_m), \quad \int_0^1 g_{\ell m}(v) dv = 0, \quad \ell = 1, \dots, p. \quad (2.2)$$

The constant c_ℓ is the level of the ℓ th coefficient and $g_{\ell m}$ is the component of that coefficient along the m th modulator. The centering in (2.2) is with respect to Lebesgue measure, which is the convention of the wavelet literature and the one the basis of Section 2.2 imposes at no cost; the parametrization centered at the law of $\mathbf{U}$ is obtained from it by the shift

2.1 The additive functional coefficient model

$c_\ell \mapsto c_\ell + \sum_m \mathbb{E}\{g_{\ell m}(U_m)\}$, which moves mass between the levels and the components but changes neither β_ℓ nor the estimator.

The model contains several familiar ones. With $X_1 \equiv 1$ the first coefficient is an additive intercept, so that $p = 1$ gives the additive model and $p > 1$ with $g_{\ell m} \equiv 0$ for $\ell \geq 2$ gives the partially linear additive model; $q = 1$ gives the classical varying coefficient model with a single index; and $g_{\ell m} \equiv 0$ for all pairs gives linear regression. Writing $f(x, u) = \sum_\ell x_\ell \beta_\ell(u)$ for the regression function, the first question is whether the distribution of $(Y, \mathbf{X}, \mathbf{U})$ determines the parameters. Two degeneracies have to be excluded: if $\mathbf{X}$ is confined to a hyperplane given $\mathbf{U}$, only a combination of the coefficients is determined, and if one modulator is a function of another, only the sum of the corresponding components is.

Proposition 1 (Identification). *Suppose that $\mathbb{E} \|\mathbf{X}\|_2^2 < \infty$, that $\mathbb{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})$ is nonsingular almost surely, and that $\mathbf{U}$ has a Lebesgue density on $[0, 1]^q$ that is positive almost everywhere. Then, among the parametrizations that satisfy (2.2), the levels $c_1, \dots, c_p$ and the components $g_{\ell m} \in L_2[0, 1]$ are determined by f : the levels uniquely, and each component up to a Lebesgue null set.*

Dependence between $\mathbf{X}$ and $\mathbf{U}$ is allowed, and so is dependence among the modulators; what the two conditions exclude is a linear combination of

the linear covariates that is a function of the modulators, and a modulator that is a function of the others. Both exclusions return in quantitative form in Section 3.

2.2 The wavelet [approximation space](#)

Let ϕ and ψ be the scaling function and the mother wavelet of an orthonormal multiresolution analysis of $L_2(\mathbb{R})$, compactly supported, with ψ bounded and with N vanishing moments, and let L be the length of the filter, so that the support of ψ is contained in $[0, L - 1]$. Write

$$\psi_{jk}(u) = \sum_{\iota \in \mathbb{Z}} 2^{j/2} \psi\{2^j(u + \iota) - k\}, \quad j \geq 0, \quad k = 0, \dots, 2^j - 1, \quad (2.3)$$

for the periodization of the basis to $[0, 1]$. The system $\{\mathbf{1}\} \cup \{\psi_{jk}\}$ is an orthonormal basis of $L_2[0, 1]$ under Lebesgue measure and every wavelet in it integrates to zero (Daubechies, 1992, Section 9.3). We take the coarsest level to be $j_0 = 0$, so that the only scaling function is the constant $\phi_{00} \equiv 1$, and we discard it: it is already carried by the level c_ℓ . The [approximation space \(the sieve\)](#) of resolution J is then

$$W_J = \text{span}\{\psi_{jk} : 0 \leq j < J, 0 \leq k < 2^j\}, \quad \dim W_J = N_J = 2^J - 1,$$

the orthogonal complement of the constants inside the multiresolution space V_J , and Π_J is the orthogonal projection onto it. Every element of W_J

integrates to zero, so that the identification constraint of (2.2) holds by construction, no numerical restriction is imposed on the fit, and the [approximation space](#) is a plain linear space.

Lemma 1 (The basis carries the constraint). *Under the conditions of Proposition 1, for every J the map*

$$(\mathbf{c}, \boldsymbol{\theta}) \mapsto \sum_{\ell=1}^p x_{\ell} \left\{ c_{\ell} + \sum_{m=1}^q \sum_{j < J} \sum_k \theta_{\ell m, jk} \psi_{jk}(u_m) \right\}$$

is injective on $\mathbb{R}^p \times \mathbb{R}^d$, $d = pqN_J$; equivalently, the population Gram matrix of the design defined in (2.4) is positive definite. If instead the column $X_{\ell}\phi_{00}(U_m)$ is retained in each block, that matrix is singular with nullity exactly pq .

Replacing $g_{\ell m}$ by $\Pi_J g_{\ell m}$ leaves a bias, which Section 3 controls under a Besov condition. Two practical consequences of periodization deserve mention. A component whose values at the two ends of the interval disagree is not periodic, and its projection error then stalls at the order $2^{-J/2}$ in L_2 regardless of the interior smoothness (Proposition S3.1 in the Supplementary Material). The two remedies, rescaling the modulators to a proper subinterval and using the basis of Cohen, Daubechies, and Vial (1993) adapted to the interval, are matters of computation and are discussed in Section 4.

2.3 The estimator

Let $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i = 1, \dots, n$, be independent copies of $(Y, \mathbf{X}, \mathbf{U})$. The row of the design matrix is the vector of the p linear covariates followed by the $d = pqN_J$ products between a linear covariate and a wavelet of a modulator,

$$\mathbf{Z} = [\mathbf{A} \ \mathbf{B}], \quad \mathbf{A}_{i\ell} = X_{i\ell}, \quad \mathbf{B}_{i,(\ell m, jk)} = X_{i\ell} \psi_{jk}(U_{im}), \quad (2.4)$$

with the blocks (ℓ, m) in lexicographic order and, inside a block, the wavelets ordered by level and then by translation. The penalty acts on chunks of wavelet coefficients inside a block. Fix an integer $b \geq 2$ and let j^* be the finest level with $2^{j^*} < b$. In each block (ℓ, m) the levels $j \leq j^*$ form a single chunk, the coarse chunk, and each finer level, which has $2^j \geq b$ coefficients, is cut into $\lfloor 2^j/b \rfloor$ chunks of consecutive translations, all with b coefficients except the last, which also takes the remainder. Write $\mathcal{G}$ for the resulting partition of the d penalized coordinates, $\boldsymbol{\theta}_G$ and $|G|$ for the subvector and the size of a chunk G , and, for positive weights $w = (w_G)_{G \in \mathcal{G}}$, $\|\boldsymbol{\theta}\|_{\mathcal{G}, w} = \sum_{G \in \mathcal{G}} w_G \|\boldsymbol{\theta}_G\|_2$. Writing $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$, the estimator is

[removed in k = 4]

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}) \in \arg \min_{(\mathbf{c}, \boldsymbol{\theta}) \in \mathbb{R}^p \times \mathbb{R}^d} \left\{ \|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_n^2 + 2\lambda \|\boldsymbol{\theta}\|_1 \right\}.$$

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}) \in \arg \min_{(\mathbf{c}, \boldsymbol{\theta}) \in \mathbb{R}^p \times \mathbb{R}^d} \left\{ \|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_n^2 + 2\lambda \|\boldsymbol{\theta}\|_{\mathcal{G}, w} \right\}, \quad (2.5)$$

with $b = b_n = \lceil \log n \rceil$ and $w_G = |G|^{1/2}$. Every chunk of a fine level has between b and $2b - 1$ coefficients, the coarse chunk has $2^{j^*+1} - 1$, which lies between $b - 1$ and $2b - 3$ (when $J \leq j^* + 1$ each block is a single chunk), there are at most $pq(1 + 2^J/b)$ chunks, and the ratio $\varrho = \max_G w_G / \min_G w_G$ of the weights satisfies $\varrho^2 \leq (2b - 1)/(b - 1) \leq 3$ (Proposition S5.1 in the Supplementary Material). The theory of Section 3 holds for any weights whose ratio stays bounded, and the bound on ϱ is what makes it cover the weights $|G|^{1/2}$ that group lasso software uses by default. The penalty does not reach the levels $\mathbf{c}$: they are p parameters of a parametric part, and shrinking them would bias every fitted coefficient by a constant. The fitted components and coefficients are recovered from $\widehat{\boldsymbol{\theta}}$ by

$$\widehat{g}_{\ell m}(v) = \sum_{j < J} \sum_k \widehat{\theta}_{\ell m, jk} \psi_{jk}(v), \quad \widehat{\beta}_{\ell}(u) = \widehat{c}_{\ell} + \sum_{m=1}^q \widehat{g}_{\ell m}(u_m).$$

Problem (2.5) is a ~~lasso~~ (Tibshirani, 1996) **group lasso** (Yuan and Lin, 2006) with some coordinates unpenalized, so it is convex, and it is strictly convex, hence uniquely solved, whenever the Gram matrix of $\mathbf{Z}$ is nonsingular. ~~Two structural variants are available at no conceptual cost and are discussed in Section 4: a sparse group lasso (Simon et al., 2013) whose groups are the~~

blocks (ℓ, m) , which removes an entire component when a modulator does not act on a coefficient, and the block lasso of Klopp and Pensky (2015). It differs from the block lasso of Klopp and Pensky (2015) in three respects: their chunks run across resolution levels, their weights are equal, and their levels are penalized with everything else. Two other penalties are available and are discussed in Section 4: the lasso (Tibshirani, 1996), with $\|\boldsymbol{\theta}\|_1$ in place of $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$, whose theory is given in Section S8 of the Supplementary Material, and a sparse group lasso (Simon et al., 2013) whose groups are the blocks (ℓ, m) . The theory below is for (2.5).

3. Theory

3.1 Assumptions

Assumption 1 (Sampling and error). The observations $(Y_i, \mathbf{X}_i, \mathbf{U}_i)$, $i \leq n$, are independent and identically distributed and follow (2.1), and the error is conditionally sub-Gaussian: there is $\sigma < \infty$ with $E\{\exp(t\varepsilon_i) \mid \mathbf{X}_i, \mathbf{U}_i\} \leq \exp(\sigma^2 t^2/2)$ almost surely, for all real t .

Assumption 2 (Linear covariates). $\max_{\ell} |X_{\ell}| \leq B_X$ almost surely, and there are constants $0 < \kappa_1 \leq \kappa_2 < \infty$ with $\kappa_1 \leq \lambda_{\min}\{E(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \lambda_{\max}\{E(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \kappa_2$ almost surely.

Assumption 3 (Modulating covariates). $\mathbf{U}$ takes values in $[0, 1]^q$ and has a joint Lebesgue density $p_{\mathbf{U}}$ with $0 < c_U \leq p_{\mathbf{U}}(u) \leq C_U < \infty$ for every $u \in [0, 1]^q$.

Assumption 4 (Basis). The wavelets are the periodization (2.3) of an orthonormal, compactly supported family with ψ bounded and N vanishing moments, and $j_0 = 0$.

Assumption 5 (Besov regularity). There are $s > 0$, $1 \leq \pi, r \leq \infty$ and $C_g < \infty$ such that every component satisfies $\int_0^1 g_{\ell m} = 0$ and, writing $\theta_{\ell m, jk}^* = \langle g_{\ell m}, \psi_{jk} \rangle$ and $\theta_{\ell m, j}^*$ for the vector of the coefficients of level j ,

$$\|\theta_{\ell m, j}^*\|_{\pi} \leq C_g 2^{-j(s+1/2-1/\pi)} \quad \text{for every } j \geq 0.$$

The effective regularity $s' = s - (1/\pi - 1/2)_+$ is positive.

Assumptions 2 and 3 are the quantitative forms of the two conditions of Proposition 1. In the second it is the joint density that is bounded below, not the marginals: if $U_2 = U_1$ almost surely both marginals are uniform and the design is singular. Assumption 5 is taken as a condition on the sequence of coefficients rather than as membership in a function space, as in Klopp and Pensky (2015); when ψ has more than s vanishing moments and enough Hölder regularity the two are equivalent, on the circle (Cohen, 2003, Section 3.6), and the implication that the proofs use needs only the

vanishing moments. The index s' is the regularity that [linear approximation](#) sees and s is the one that nonlinear approximation sees; they differ exactly when $\pi < 2$, which is the regime of inhomogeneous smoothness.

3.2 The main result

Write $\widehat{f}_i = \mathbf{A}_i \widehat{\mathbf{c}} + \mathbf{B}_i \widehat{\boldsymbol{\theta}}$ for the fit, ~~d_n for the number of penalized columns~~ at resolution ~~$J_n \mathcal{G}_n$ for the partition of Section 2.3~~ at resolution J_n with $b = b_n$, and fix a confidence level $\alpha \in (0, 1)$. The penalty is the one that the oracle inequality of Section 3.4 calls for, namely, with $\gamma = \kappa_1 C_U$ and $\Lambda = \kappa_2 C_U + \gamma/2$ the constants of Proposition 2 below,

[removed in k = 4]

$$\lambda_n = 2\sigma B_X \sqrt{2C_U} \sqrt{\frac{2 \log(2d_n/\alpha)}{n}} \asymp \sigma B_X \sqrt{C_U} \sqrt{\frac{J_n}{n}}.$$

$$\lambda_n^{\mathcal{G}} = 2 \max_{G \in \mathcal{G}_n} \frac{\bar{\lambda}_{0,G}}{w_G}, \quad \bar{\lambda}_{0,G} = \frac{\sigma}{\sqrt{n}} \left\{ B_X \sqrt{2C_U |G|} + \sqrt{2\Lambda \log(|\mathcal{G}_n|/\alpha)} \right\}. \quad (3.1)$$

For every chunk, $\lambda_n^{\mathcal{G}} w_G \leq \varrho \lambda_{n,1}^{\mathcal{G}}$, where $\lambda_{n,1}^{\mathcal{G}} = 2 \max_G \bar{\lambda}_{0,G}$ is of order $\sigma(\log n/n)^{1/2}$, the order of the penalty of the lasso.

[removed in k = 4]

Theorem (Rate of nonlinear approximation). *Let Assumptions 1 to 5 hold*

3.2 The main result

with p, q and all the constants fixed as n grows, and set $\tau = (s + 1/2)^{-1}$.

Suppose $s' > s/(2s + 1)$ and take $J_n = \lceil c \log_2 n \rceil$ with

$$\max\left\{\frac{\tau}{2}, \frac{2 - \tau}{4s'}\right\} \leq c < 1,$$

a window that is nonempty precisely under that condition. Then, with λ_n

as in the display above,

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}, \quad \sum_{\ell, m} \left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s/(2s+1)}\right\}.$$

Theorem 1 (Rate of nonlinear approximation). *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and let the weights have ratio $\varrho \leq \bar{\varrho}$ for a constant $\bar{\varrho}$ fixed as n grows ($\bar{\varrho}^2 = 3$ for the weights of Section 2.3). Suppose $s' > s/(2s + 1)$ and take $J_n = \lceil c \log_2 n \rceil$ with*

$$\frac{s}{(2s + 1)s'} \leq c < 1, \tag{3.2}$$

a window that is nonempty precisely under that condition. Then, with $\lambda =$

$\lambda_n^{\mathcal{G}}$ as in (3.1) and $r_n = n^{-2s/(2s+1)}(\log n)^{(2/\pi-1)/(2s+1)}$,

$$\left\|\widehat{f} - f\right\|_n^2 = O_p(r_n), \quad \sum_{\ell, m} \left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 = O_p(r_n).$$

In particular, when $\pi \geq 2$ the rate is $n^{-2s/(2s+1)}$, with no logarithmic factor.

Three readings of Theorem 1 matter. First, the exponent is the one of nonlinear wavelet approximation, $2s/(2s + 1)$, and not the one of linear approximation, $2s'/(2s' + 1)$ of Theorem 3 below; the two agree when

$\pi \geq 2$ and separate when $\pi < 2$, which is where the ℓ_1 penalty pays for itself. Second, the rate is attained under Assumption 5 alone: ~~no~~ compressibility is assumed on top of it, because the Besov condition already implies it, with $\tau = (s + 1/2)^{-1}$, by Lemma 3. ~~no condition is imposed on the coefficients on top of it, because the Besov condition already bounds the ideal risk over chunks that governs the estimation error (Lemma 3).~~ Third, the rate $(\log n/n)^{2s/(2s+1)}$ is the minimax reference of the Gaussian sequence model over Besov bodies with per coordinate noise of order λ_n (Donoho and Johnstone, 1998, Theorems 4 and 5; Johnstone, 2019, Theorem 9.3); the logarithmic factor is the price of the single, nonadaptive penalty level (3.1), which takes a union bound over the d_n columns. No lower bound is proved here for $q \geq 2$: the comparison is with the reference of the sequence model, and the minimax lower bound of Klopp and Pensky (2015) covers the case of a single modulator with covariates independent of it. **Third,** $n^{-2s/(2s+1)}$ is the minimax rate over Besov bodies in the Gaussian sequence model with noise level of order $n^{-1/2}$ (Donoho and Johnstone, 1998, Theorems 4 and 5), and with a single modulator, independent of the covariates, and Gaussian errors, the minimax lower bound of Klopp and Pensky (2015, Theorem 1) holds over the class of Assumption 5 (Remark S6.1 in the Supplementary Material), so that in that case Theorem 1 is minimax optimal

when $\pi \geq 2$, the rate cannot be improved. With Gaussian errors and any law of $(\mathbf{X}, \mathbf{U})$ that satisfies Assumptions 2 and 3, the minimax risk over the class of Assumption 5, in the prediction norm and for the components, is at least a constant multiple of $pq n^{-2s/(2s+1)}$, for every π (Theorem S6.1 in the Supplementary Material). This is the regression counterpart of the lower bound of Donoho and Johnstone (1998, Theorem 5) for Besov bodies in the Gaussian sequence model, and with a fixed number of covariates it extends that of Klopp and Pensky (2015, Theorem 1), who have a single modulator independent of the covariates. Theorem 1 is therefore minimax optimal when $\pi \geq 2$, with any number of modulators and with covariates that depend on them. When $\pi < 2$ the remaining factor $(\log n)^{(2/\pi-1)/(2s+1)}$ is the one in the rate of Klopp and Pensky (2015, eq. 3.19) with a fixed number of covariates, and in that of block thresholding in the sequence model (Cai, 1999, Theorem 4); whether it is necessary for this estimator is not known, and no lower bound is proved here for $q \geq 2$. the lower bound carries no such factor, and whether the factor is necessary for this estimator is not known. Compared with the coordinatewise lasso, whose rate under the same conditions is $(\log n/n)^{2s/(2s+1)}$ (Theorem S8.3 in the Supplementary Material), the chunks remove a factor $(\log n)^{2s'/(2s+1)}$. When $s < 1/2$ and $\pi \geq 2$, the rate $n^{-2s/(2s+1)}$ of Theorem 1 is attained with no design condition

3.3 Approximation and the design condition

at all, at another resolution (Proposition 3(ii) below). When $s < 1/2$, the prediction rate of Theorem 1 holds with no design condition at all, under its other assumptions and in the same window for J_n (Proposition S6.2 in the Supplementary Material).

The proof combines three ingredients, given below: the approximation error of W_J , the design condition, and a nonasymptotic oracle inequality. Section 3.5 assembles them, first at a dense reading of the estimation term the oracle, which gives the rate of linear approximation, and then at a sparse-comparator a comparator that keeps only the chunks of large energy, which gives Theorem 1.

3.3 Approximation and the design condition

Lemma 2 (Approximation bias). *Under Assumptions 4 and 5, for every pair (ℓ, m) and every $J \geq 0$,*

$$\|g_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{j \geq J} \|\theta_{\ell m, j}^*\|_2^2 \leq \frac{C_g^2}{1 - 2^{-2s'}} 2^{-2Js'}.$$

If in addition Assumptions 2 and 3 hold, then, with $f_J(x, u) = \sum_{\ell} x_{\ell} \{c_{\ell} + \sum_m (\Pi_J g_{\ell m})(u_m)\}$,

$$\|f - f_J\|_{L_2(P)}^2 \leq \frac{(pq)^2 B_X^2 C_U C_g^2}{1 - 2^{-2s'}} 2^{-2Js'} =: \mathcal{B}_J^2. \quad (3.3)$$

The exponent is s' and not s , and this is where the loss $(1/\pi - 1/2)_+$

3.3 Approximation and the design condition

enters: inside a level, an ℓ_π bound with $\pi < 2$ controls the ℓ_2 norm without gain. The bias uses only the bound on the linear covariates and the marginal density of each modulator, so dependence between $\mathbf{X}$ and $\mathbf{U}$ does not affect it. What it does affect is the passage from the prediction norm to the norm of each component, which is the design condition.

Let $\Psi(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, collect the constant and the wavelets of the q modulators, so that the row of $\mathbf{Z}$ in (2.4) is a permutation of the Kronecker product $\mathbf{X} \otimes \Psi(\mathbf{U})$. Write $\Sigma = \mathbb{E}(\mathbf{Z}\mathbf{Z}^\top)$ and $\widehat{\Sigma} = \mathbf{Z}^\top \mathbf{Z}/n$ for the population and empirical Gram matrices and $\Sigma_\Psi = \mathbb{E}\{\Psi(\mathbf{U})\Psi(\mathbf{U})^\top\}$ for the part in the modulators alone.

Proposition 2 (Gram matrix of the product design). *Let Assumptions 2 to 4 hold and write $D = p + d$ and $\gamma = \kappa_1 c_U$.*

(i) $c_U \leq \lambda_{\min}(\Sigma_\Psi) \leq \lambda_{\max}(\Sigma_\Psi) \leq C_U$ for every J , and consequently $\gamma \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq \kappa_2 C_U$. If $\mathbf{X}$ and $\mathbf{U}$ are independent, Σ is, up to the permutation of columns, the Kronecker product $\mathbb{E}(\mathbf{X}\mathbf{X}^\top) \otimes \Sigma_\Psi$, and the two bounds are attained as products of eigenvalues.

(ii) With $R_J = pB_X^2(1 + qC_\psi 2^J)$, where C_ψ depends only on the wavelet family, $\|\mathbf{Z}_i\|_2^2 \leq R_J$ almost surely and

$$\mathbb{P}\left(\lambda_{\min}(\widehat{\Sigma}) \geq \frac{\gamma}{2}\right) \geq 1 - 2D \exp\left\{-\frac{n\gamma^2}{8R_J(\kappa_2 C_U + \gamma/3)}\right\},$$

which tends to one whenever $pq2^J \log(pq2^J)/n \rightarrow 0$.

Part (i) is stated for the general case: independence between the covariates and the modulators buys the exact factorization but not a better constant. A lower bound on the smallest eigenvalue dominates every compatibility and restricted eigenvalue constant, including the versions for groups of Lounici et al. (2011), on every support and with no cone restriction, since $\delta^\top A \delta \geq \lambda_{\min}(A) \|\delta\|_2^2 \geq \lambda_{\min}(A) \|\delta_S\|_1^2 / |S|$, and likewise $\delta^\top A \delta \geq \lambda_{\min}(A) \{\sum_{G \in \mathcal{H}} w_G \|\delta_G\|_2\}^2 / \mathcal{W}(\mathcal{H})$ for a set $\mathcal{H}$ of chunks, with $\mathcal{W}(\mathcal{H}) = \sum_{G \in \mathcal{H}} w_G^2$, and it is in this form that the next subsection consumes it. In the resolution regime of Section 3.5 the requirement of part (ii) is of the same order as the rate itself the rate times $\log n$, which tends to zero, so it costs nothing there; if pq were larger than $n^{1/2}$, only the compatibility constant could be transferred, through the entrywise deviation (Bühlmann and van de Geer, 2011, Corollary 6.8), at the price of a factor s_0^2 and of the cone.

3.4 A nonasymptotic oracle inequality

Let $\beta^* = (\mathbf{c}, \boldsymbol{\theta}^*)$ be the oracle of the approximation space, that is the vector whose components are the levels and the coefficients $\theta_{\ell m, jk}^*$ of Assumption 5 for $j < J$, so that f_J is its fit, and let

3.4 A nonasymptotic oracle inequality

[removed in k = 4]

$$\mathbf{b} = f - f_J \in \mathbb{R}^n, \quad S_0 = \text{supp}(\boldsymbol{\theta}^*), \quad s_0 = |S_0|, \quad v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*.$$

$\mathbf{b} = f - f_J \in \mathbb{R}^n$, $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$, $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$, $\bar{\mathcal{G}} = \{G : \bar{\boldsymbol{\theta}}_G \neq 0\}$,
for a comparator $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}) \in \mathbb{R}^p \times \mathbb{R}^d$, any vector of levels and coefficients,
 $\bar{\mathcal{G}}$ being its active chunks, and $\bar{\mathcal{W}} = \mathcal{W}(\bar{\mathcal{G}}) = \sum_{G \in \bar{\mathcal{G}}} w_G^2$. Let $\Pi_{\mathbf{A}}$ be the
orthogonal projection onto the column space of $\mathbf{A}$, which exists whenever
 $\mathbf{A}$ has rank p , let $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$, and let $\tilde{\mathbf{B}} = M_{\mathbf{A}}\mathbf{B}$ and $\tilde{\gamma} = \lambda_{\min}(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}/n)$
be the residualized design and its smallest eigenvalue, and $\tilde{\Psi}_G = \tilde{\mathbf{B}}_G^\top \tilde{\mathbf{B}}_G/n$
the residualized Gram matrix of the columns of a chunk. Finally let $\widehat{\sigma}_{\max}^2 =$
 $\max_a (\widehat{\Sigma})_{aa}$ be the largest empirical norm of a penalized column.

[removed in k = 4]

Theorem (Oracle inequality). *Let Assumption 1 hold, suppose $\mathbf{A}$ has rank p , and take $\lambda \geq 2\lambda_0$ with $\lambda_0 = \sigma \widehat{\sigma}_{\max} \{2 \log(2d/\alpha)/n\}^{1/2}$. On the event $\mathcal{T} = \{\|\tilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\|_\infty \leq \lambda/2\}$, which has probability at least $1 - \alpha$:*

$$(i) \text{ with no condition on the design, } \left\| \tilde{\mathbf{B}}v \right\|_n^2 \leq 6\lambda \|\boldsymbol{\theta}^*\|_1 + 4 \|\mathbf{b}\|_n^2;$$

(ii) if $\tilde{\gamma} > 0$,

$$\left\| \tilde{\mathbf{B}}v \right\|_n^2 \leq \frac{64\lambda^2 s_0}{\tilde{\gamma}} + 16 \|\mathbf{b}\|_n^2, \quad \|v\|_2^2 \leq \frac{64\lambda^2 s_0}{\tilde{\gamma}^2} + \frac{16 \|\mathbf{b}\|_n^2}{\tilde{\gamma}};$$

3.4 A nonasymptotic oracle inequality

(iii) in either regime, $\left\|\widehat{f} - f\right\|_n \leq \left\|\widetilde{\mathbf{B}}v\right\|_n + 2\left\|\mathbf{b}\right\|_n + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n$, and $\mathbb{E}(\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$.

Theorem 2 (Oracle inequality). *Let Assumption 1 hold, suppose $\mathbf{A}$ has rank p , let the weights not depend on the responses, and, for $\alpha \in (0, 1)$, let*

$$\lambda_{0,G} = \frac{\sigma}{\sqrt{n}} \left[\{\text{tr}(\widetilde{\Psi}_G)\}^{1/2} + \{2 \left\|\widetilde{\Psi}_G\right\|_{\text{op}} \log(|\mathcal{G}|/\alpha)\}^{1/2} \right]$$

and $\lambda_w = 2 \max_{G \in \mathcal{G}} \lambda_{0,G}/w_G$, and take $\lambda \geq \lambda_w$. On the event $\mathcal{T}_{\mathcal{G},w} = \{\max_G \left\|(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G\right\|_2 / w_G \leq \lambda/2\}$, which has probability at least $1 - \alpha$, the following hold simultaneously for every comparator $\bar{\boldsymbol{\beta}}$:

(i) with no condition on the design, $\frac{1}{2} \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + \lambda \|\widehat{\boldsymbol{\theta}}\|_{\mathcal{G},w} \leq 3\lambda \|\bar{\boldsymbol{\theta}}\|_{\mathcal{G},w} + 2 \left\|\bar{\mathbf{b}}\right\|_n^2$;

(ii) if $\widetilde{\gamma} > 0$,

$$\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \leq \frac{64\lambda^2\bar{\mathcal{W}}}{\widetilde{\gamma}} + 16 \left\|\bar{\mathbf{b}}\right\|_n^2, \quad \|\bar{v}\|_2^2 \leq \frac{64\lambda^2\bar{\mathcal{W}}}{\widetilde{\gamma}^2} + \frac{16 \left\|\bar{\mathbf{b}}\right\|_n^2}{\widetilde{\gamma}};$$

(iii) in either regime, $\left\|\widehat{f} - f\right\|_n \leq \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n + 2 \left\|\bar{\mathbf{b}}\right\|_n + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n$, and $\mathbb{E}(\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$.

ThreeFour features separate Theorem 2 from the standard treatment (Bühlmann and van de Geer, 2011, Chapter 6) and from that of the group

lasso by Lounici et al. (2011). The unpenalized levels are not placed inside the support, which would require a compatibility condition on a cone; they are profiled out, and the Schur complement transfers the full smallest eigenvalue of $\widehat{\Sigma}$ to the residualized block, $\tilde{\gamma} \geq \lambda_{\min}(\widehat{\Sigma})$, so that Proposition 2(ii) is used as it stands and the constant does not depend on $\mathcal{S}_0 \bar{\mathcal{G}}$. The model is misspecified inside the approximation space, and the deterministic residual $\mathfrak{h} \bar{\mathbf{b}}$ is handled by Young's inequality rather than by duality, which keeps the statement nonasymptotic and valid also when the bias dominates, and the statement holds for every comparator at once, which is what Section 3.5 uses. The penalty is calibrated by the empirical column norms and not by $\max_{i,a} |Z_{ia}|$: the former is bounded in probability by $2B_X^2 C_U$, while the latter is of order $2^{J/2}$ and would inflate λ by that factor and destroy the rates of Section 3.5. The penalty is calibrated chunk by chunk, by the tail inequality for quadratic forms of Hsu et al. (2012, Theorem 2.1) with a union bound over the chunks (Lemma S5.2 in the Supplementary Material); the event $\mathcal{T}_{\mathcal{G},w}$ does not depend on the weights, which only choose the chunk that dictates λ_w , and the price of unequal weights is the factor ϱ in $\lambda_w w_G \leq 2\varrho \max_{G'} \lambda_{0,G'}$, paid in the estimation term. Since $\text{tr}(\tilde{\Psi}_G) \leq |G| \hat{\sigma}_{\max}^2$ and $\left\| \tilde{\Psi}_G \right\|_{\text{op}} \leq \lambda_{\max}(\widehat{\Sigma})$, the deterministic level (3.1) dominates λ_w whenever $\hat{\sigma}_{\max}^2 \leq 2B_X^2 C_U$ and $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$, two

events of probability tending to one in the regime of Section 3.5; it is the empirical column norms that enter, and not $\max_{i,a} |Z_{ia}|$, which is of order $2^{J/2}$ and would destroy the rates. The software standardizes the columns and orthonormalizes each chunk before penalizing it; the theorem covers that estimator as well, with $\tilde{\Psi}_G$ replaced by $Q_G^{-1/2} \tilde{\Psi}_G Q_G^{-1/2}$, Q_G being the centered Gram matrix of the chunk, which makes $\lambda_{0,G}$ pivotal, and with $\bar{\mathcal{W}}$ multiplied by $\max_G \lambda_{\max}(Q_G)$ (Remark S5.2 in the Supplementary Material). The term $\sigma^2 p/n$ in part (iii) is the price of leaving the p levels out of the penalty; no choice of λ removes it, it is of smaller order than the estimation term whenever $p \lesssim s_0 \log(pqN_J) p \lesssim |\bar{\mathcal{G}}| \log n$, and it is absent from Klopp and Pensky (2015), where the levels are penalized along with everything else.

3.5 Rates

Under Assumption 5 the [oracle](#) is generically dense: the coefficients decay but do not vanish, so that $s_0 = d$ [every chunk is active](#) and part (ii) of Theorem 2, read at the oracle, delivers the rate of [linear approximation](#), [now with no logarithmic factor](#).

[removed in k = 4]

Theorem (Rate of linear approximation). *Let Assumptions 1 to 5 hold*

with p, q and the constants fixed, and take

$$J_n = \min\{J \in \mathbb{N} : 2^J \geq (n/\log n)^{1/(2s'+1)}\},$$

so that $J_n = \log_2 n / (2s' + 1) + O(\log \log n)$. Then the requirement of Proposition 2(ii) holds along J_n , being itself of the exact order $(\log n/n)^{2s'/(2s'+1)}$, and, with λ_n as in the first display of Section 3.2,

$$\|\widehat{f} - f\|_n^2 = O_p\left\{\left(\frac{\log n}{n}\right)^{2s'/(2s'+1)}\right\}.$$

At this resolution the estimation term and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem 2(iii) is of strictly smaller order.

Theorem 3 (Rate of linear approximation). *Let Assumptions 1 to 5 hold with p, q and the constants fixed, let the weights be as in Theorem 1, and take*

$$J_n = \min\{J \in \mathbb{N} : 2^J \geq n^{1/(2s'+1)}\}. \quad (3.4)$$

Then the requirement of Proposition 2(ii) holds along J_n , being of order $n^{-2s'/(2s'+1)} \log n$, and, with $\lambda = \lambda_n^{\mathcal{G}}$ as in (3.1),

$$\|\widehat{f} - f\|_n^2 = O_p(n^{-2s'/(2s'+1)}).$$

At this resolution the estimation term, of order $pq(b_n + 2^{J_n})/n$, and the bias term are of the same order, and the term $\sigma^2 p/n$ of Theorem 2(iii) is of strictly smaller order.

[removed in k = 4]

Corollary (Components and coefficients). *In the setting of Theorem 3, and with r_n denoting the rate of that theorem,*

$$\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p(r_n), \quad \|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} = O_p(r_n^{1/2}), \quad \ell = 1, \dots, p,$$

and the same statements hold with the rate of Theorem 1 in place of r_n under the conditions of that theorem.

Corollary 1 (Components and coefficients). *In the setting of Theorem 3,*

with $\rho_n^{\mathcal{G}} = n^{-s'/(2s'+1)}$,

$$\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p\{(\rho_n^{\mathcal{G}})^2\}, \quad \|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} = O_p(\rho_n^{\mathcal{G}})$$

for $\ell = 1, \dots, p$, and $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 = O_p(\rho_n^{\mathcal{G}})$.

The passage from the prediction norm to the norm of a single component is where the design condition is used a second time, and the smallest eigenvalue appears squared in the constant. In the arguments of this paper the design condition is needed for the components in every case, and for the prediction rate of Theorem 1 only when $s \geq 1/2$. Two further statements complete the picture. The first dispenses with the design condition altogether, at the price of a slower rate when $s \geq 1/2$, and is the unconditional statement of this paper.

[removed in k = 4]

Proposition (Slow rate, no design condition). *Let Assumptions 1, 4 and 5 hold, with $\mathbf{A}$ of rank p and λ_n as in the first display of Section 3.2. Then $\left\|\widehat{f} - f\right\|_n^2 = O_p\{\lambda_n \|\boldsymbol{\theta}^*\|_1 + \mathcal{B}_{J_n}^2 + p/n\}$, where $\|\boldsymbol{\theta}^*\|_1 \leq pqC_g \sum_{j < J_n} 2^{j(1/2-s)}$ does not depend on π . In particular, if $s > 1/2$ then $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(J_n/n)^{1/2}\}$ for every $J_n \geq \log_2 n/(4s')$; and if $s < 1/2$, the choice $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ gives $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(J_n/n)^{2s'/(1-2s+4s')}\}$, an exponent that equals $2s/(2s+1)$ when $\pi \geq 2$.*

Proposition 3 (Slow rate, no design condition). *Let Assumptions 1, 4 and 5 hold, together with the upper bounds in Assumptions 2 and 3 ($\max_\ell |X_\ell| \leq B_X$, $\lambda_{\max}\{\mathbf{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\} \leq \kappa_2$ and $p_{\mathbf{U}} \leq C_U$), with $\mathbf{A}$ of rank p , the weights as in Theorem 1, $\lambda = \lambda_n^{\mathcal{G}}$ as in (3.1) with any constant $\Lambda > \kappa_2 C_U$, $2^{J_n} \leq 2n$ and $pq2^{J_n} \log(pq2^{J_n})/n \rightarrow 0$. Then*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + \mathcal{B}_{J_n}^2 + p/n\},$$

with $\lambda_n^{\mathcal{G}} \|\boldsymbol{\theta}^\|_{\mathcal{G},w} \leq \bar{\varrho} \lambda_{n,1}^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \leq \bar{\varrho} \lambda_{n,1}^{\mathcal{G}} \|\boldsymbol{\theta}^*\|_1$, where $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = \sum_G \|\boldsymbol{\theta}_G^*\|_2$ and $(\lambda_{n,1}^{\mathcal{G}})^2$ is of order $\sigma^2 \log n/n$. Write $\vartheta_\pi = \min(1 - 1/\pi, 1/2)$.*

(i) *If $s > 1/2$ and $s' > 1/4$, then $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O(1)$ and $\left\|\widehat{f} - f\right\|_n^2 = O_p\{(\log n/n)^{1/2}\}$ for every $J_n \geq \log_2 n/(4s')$ that satisfies the conditions above, and one does.*

(ii) If $s < 1/2$ and $s' > s/2$, then $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O\{1 + b_n^{-\vartheta_\pi} 2^{J_n(1/2-s)}\}$, and the choice $2^{J_n} \asymp \{n/b_n^{(2/\pi-1)+}\}^{1/(1-2s+4s')}$ gives

$$\left\| \widehat{f} - f \right\|_n^2 = O_p\{n^{-2s'/(1-2s+4s')} (\log n)^{(2/\pi-1)+2s'/(1-2s+4s')}\},$$

which, when $\pi \geq 2$, is $n^{-2s/(2s+1)}$, the rate of Theorems 1 and 3, with no logarithmic factor.

The proof uses no lower bound on $\widehat{\Sigma}$, and no property of $\widehat{\Sigma}$ other than $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$ an upper bound on $\lambda_{\max}(\widehat{\Sigma})$.

The bound on $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$ is Lemma S6.2 in the Supplementary Material. Through the chain to $\|\boldsymbol{\theta}^*\|_1$, Proposition 3 reproduces the slow rate of the lasso (Proposition S8.1), with the constant $\bar{\varrho}$: a chunk of up to $2b_n - 1$ coefficients enters $\lambda_{n,1}^{\mathcal{G}}$ at the order of the union bound over the chunks. What the chain throws away is the count: at the fine levels $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$ is smaller than $\|\boldsymbol{\theta}^*\|_1$ by the factor $b_n^{-\vartheta_\pi}$, which in (ii) cancels the factor $b_n^{1/2}$ that the size of the chunks puts in $\lambda_{n,1}^{\mathcal{G}}$, entirely when $\pi \geq 2$ and in part when $\pi < 2$; in (i) the coarse chunk dominates both norms and nothing changes. No lower bound on $\widehat{\Sigma}$ enters: the only property of $\widehat{\Sigma}$ that the proof uses is $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$ an upper bound on $\lambda_{\max}(\widehat{\Sigma})$, here $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$, which makes the deterministic level (3.1) admissible and holds with probability tending to one under the upper bounds of Assumptions 2 and 3 alone, so that neither

c_U nor κ_1 is needed. The conditions $s' > 1/4$ in (i) and $s' > s/2$ in (ii) bite only when $\pi < 2$, and make the resolution of the balance admissible. The pairs that these conditions leave out, $s' \leq 1/4$ with $s > 1/2$ and $s' \leq s/2$ with $s < 1/2$, all with $\pi < 2$, and the boundary $s = 1/2$ call for a resolution above $n/\log n$, where the column norms and $\lambda_{\max}(\widehat{\Sigma})$ grow with $2^{J_n}/n$; Proposition S6.1 in the Supplementary Material covers them with the level inflated to $\lambda_n^{\mathcal{G},+} = (\mu_{J_n}^{\mathcal{G}})^{1/2} \lambda_n^{\mathcal{G}}$, where $\mu_J^{\mathcal{G}} = 1 + c_\mu 2^J \log(2D)/n$ for a constant c_μ given there, so that the first display of Proposition 3, with $\lambda_n^{\mathcal{G},+}$ in place of $\lambda_n^{\mathcal{G}}$, holds for every $s' > 0$. There, too, the only property of $\widehat{\Sigma}$ that the proof uses is an upper bound, $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda \mu_{J_n}^{\mathcal{G}}$, which holds with probability tending to one with no condition on J_n and, again, without c_U or κ_1 . When $s < 1/2$ the slow rate improves further. Read at the comparator that keeps the chunks with $\|\theta_G^*\|_2 > \lambda w_G$ and sets the others to zero, Theorem 2(i) charges the penalty only for the chunks kept and leaves the energy of the others to the bias, where an upper bound on $\lambda_{\max}(\Sigma)$ controls it; with $s' > s/(2s+1)$ and J_n in the window (3.2), this gives the prediction rate of Theorem 1 with no design condition (Proposition S6.2 in the Supplementary Material). When $s \geq 1/2$ no comparator does better than the order $\lambda_{n,1}^{\mathcal{G}} \asymp (\log n/n)^{1/2}$ of part (i): a coarse coefficient of fixed size costs its product with $\lambda_{n,1}^{\mathcal{G}}$ in the penalty or its square in the bias.

Neither reading reaches the components, which need the design condition (Corollary 1).

The second is the approximation lemma that produces Theorem 1. For a vector $\boldsymbol{\theta}$ let $\theta_{(1)} \geq \theta_{(2)} \geq \dots$ be the nonincreasing rearrangement of the absolute values of its entries and let $\boldsymbol{\theta}^{(M)}$ retain the M largest of them. For $\boldsymbol{\theta} \in \mathbb{R}^d$ and $\eta > 0$, let $R_{\mathcal{G}}(\boldsymbol{\theta}; \eta) = \sum_{G \in \mathcal{G}} \min(\|\boldsymbol{\theta}_G\|_2^2, \eta)$ be the ideal risk over chunks: the risk of an oracle that keeps a chunk when its energy exceeds η , at the price η , and drops it otherwise.

[removed in k = 4]

Lemma (Besov regularity implies compressibility). *Under Assumptions 4 and 5 with $\pi(s+1/2) > 1$, the oracle satisfies, for every J and every $k \geq 1$,*

$$\theta_{(k)}^* \leq C_{\tau} k^{-1/\tau}, \quad \tau = \frac{1}{s+1/2}, \quad C_{\tau} = C_g(pq A_{s,\pi})^{s+1/2}, \quad A_{s,\pi} = 1 + \frac{1}{1 - 2^{1-\pi(s+1/2)}},$$

with constants that do not depend on J and with $\tau \in (0, 2)$ for every $s > 0$.

Consequently $\sum_{k>M} \theta_{(k)}^{*2} \leq \tau(2-\tau)^{-1} C_{\tau}^2 M^{1-2/\tau}$ for every $M \geq 1$.

Lemma 3 (Besov regularity bounds the ideal risk over chunks). *Under Assumptions 4 and 5, let $b \geq 1$ and let $\mathcal{G}$ be a partition in which, in each block, the levels with $2^j < b$ form at most one chunk and every level with $2^j \geq b$ is cut into chunks contained in it with at least b coefficients each, as*

the partition of Section 2.3 is. Then, for every J and every $\eta > 0$,

$$R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) \leq pq \left\{ A_{s,\pi}^{\mathcal{G}} C_g^{\tau} \left(\frac{\eta}{b} \right)^{2s/(2s+1)} b^{(2/\pi-1)_+/(2s+1)} + \eta \right\}, \quad \tau = \frac{1}{s+1/2},$$

with $A_{s,\pi}^{\mathcal{G}} = 2 + \{1 - 2^{1-\pi(s+1/2)}\}^{-1}$ if $\pi \leq 2$ and $A_{s,\pi}^{\mathcal{G}} = 2 + (1 - 2^{-2s})^{-1}$ if $\pi \geq 2$.

The inclusion of Besov balls in weak ℓ_{τ} balls is classical (Johnstone, 2019, Proposition 9.15); what the lemma adds is the constant C_{τ} , uniform in J , for the oracle of the product design, which is the form in which the proof of Theorem 1 uses it. The exponent is the one of Lemma 4 in the supplementary material of Klopp and Pensky (2015), but the assumption is weaker: their class is a weighted ℓ_{ν} ball over all the coefficients, which implies Assumption 5 level by level, and Assumption 5 does not need the sum over the levels. The coarse chunk costs at most one η per block.

Theorem 1 follows by reading Theorem 2(ii) not at the oracle but at the best M term approximation of it, which is legitimate because the proof of that theorem uses the comparator only through the decomposition of $\mathbf{Y}$ and the size of its support. Balancing the two terms that result, $\lambda^2 M / \tilde{\gamma}$ from estimation and $M^{1-2/\tau}$ from the truncation, gives the effective dimension [removed in k = 4]

$$M_n^* \asymp \left(\frac{n}{J_n}\right)^{\tau/2} \ll d_n \asymp pq 2^{J_n},$$

and a bound of order $\lambda_n^{2-\tau}$, which is the rate of Theorem 1. The three inequalities that define the window (3.2) are, in order, that the minimum in M be interior, that the approximation bias not dominate, and that the requirement of Proposition 2(ii) hold. Nothing in this argument says that the estimator selects M_n^* coordinates: no minimal signal or irrepresentability condition is used, and what compressibility buys is a rate governed by M_n^* rather than by d_n , not the recovery of a support. Theorem 1 follows by reading Theorem 2(iii) not at the oracle but at the comparator that keeps the chunks of $\boldsymbol{\theta}^*$ whose squared norm exceeds a level η and sets the others to zero, which the theorem allows because it holds for every comparator at once. The estimation term then counts η for each chunk kept, and the truncation leaves the energy of the others, so that the bound is of the order of $R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta)$ (Corollary S5.1 in the Supplementary Material), with η of the order of $\max_G(\lambda w_G)^2$, that is of b_n/n . Lemma 3 with $b = b_n$ turns this into $n^{-2s/(2s+1)} b_n^{(2/\pi-1)/(2s+1)}$: the chunk pays b_n in the level η and returns it through $(\eta/b)^{2s/(2s+1)}$, so that the cost is of order $1/n$ per coefficient, against $\log n/n$ for the lasso, and the logarithm of the union bound over the columns is absorbed by the size of the chunks, which is why they must have at least of

3.6 Which modulators act on which coefficients

the order of $\log n$ coefficients. When $\pi < 2$ the bound charges $b_n^{(2/\pi-1)/(2s+1)}$ to sequences whose large coefficients sit one per chunk; a signal spread over a level pays nothing for the chunks. The two inequalities that define the window (3.2) are that the approximation bias not dominate and that the requirement of Proposition 2(ii) hold. Nothing in this argument says that the estimator selects the chunks kept by the comparator: no minimal signal or irrepresentability condition is used. The recovery that matters for the model is not of coordinates but of blocks, and it is taken up next.

3.6 Which modulators act on which coefficients

The structure of the model is the set of blocks whose component does not vanish,

$$\mathcal{S} = \{(\ell, m) : \nu_{\ell m} > 0\}, \quad \nu_{\ell m} = \|g_{\ell m}\|_{L_2[0,1]},$$

and to know it is to know which modulators act on which coefficients. The ~~lasso~~ estimator (2.5) is not claimed to recover it: its penalty removes chunks and not blocks, under Assumption 5 the oracle is dense, and nothing prevents the fit from keeping nonzero coefficients in blocks whose component is zero. What the preceding results do give is that every block is estimated at the rate of Corollary 1, and that is enough to recover $\mathcal{S}$ in a second step, by a threshold on the norms of the fitted components. The procedure is

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estimation followed by a threshold. By the orthonormality of the basis, the norm of a fitted component is the Euclidean norm of its block of coefficients,

$$\widehat{\nu}_{\ell m} = \|\widehat{g}_{\ell m}\|_{L_2[0,1]} = \left\| \widehat{\boldsymbol{\theta}}_{\ell m, \cdot} \right\|_2, \quad \widehat{\mathcal{S}}(t) = \{(\ell, m) : \widehat{\nu}_{\ell m} > t\},$$

so that the statistic costs nothing once the estimator has been fit. Write

$$\rho_n = (\log n/n)^{s'/(2s'+1)} \rho_n^{\mathcal{G}} = n^{-s'/(2s'+1)} \text{ for the square root of the rate of}$$

Theorem 3.

Assumption 6 (Block separation). The components may depend on n , within the class of Assumption 5 and with its constants fixed, and so may $\mathcal{S}$; the smallest active block, $\nu_{\min, n} = \min_{(\ell, m) \in \mathcal{S}} \nu_{\ell m}$, with $\nu_{\min, n} = \infty$ when $\mathcal{S}$ is empty, satisfies $\nu_{\min, n}/\rho_n \rightarrow \infty$ $\nu_{\min, n}/\rho_n^{\mathcal{G}} \rightarrow \infty$.

Corollary 2 (Block structure by thresholding). *Let the assumptions of Theorem 3 hold, with J_n and $\lambda_n = \lambda_n^{\mathcal{G}}$ as there.*

(i) (Screening.) *If the thresholds satisfy $t_n/\rho_n \rightarrow \infty$ $t_n/\rho_n^{\mathcal{G}} \rightarrow \infty$, then*

$$\mathbb{P}\{\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}\} \rightarrow 1.$$

(ii) (Recovery.) *If, in addition, Assumption 6 holds and $\limsup_n t_n/\nu_{\min, n} <$*

1, then $\mathbb{P}\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \rightarrow 1$. Such thresholds always exist; one is

$$t_n = (\rho_n \nu_{\min, n})^{1/2} \quad t_n = (\rho_n^{\mathcal{G}} \nu_{\min, n})^{1/2}.$$

(iii) (A window at fixed n .) *With α the level in (3.1), for every n and every $\alpha' \in (0, 1)$ there is an event, of probability at least $1 - \alpha -$*

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$\alpha' - \omega_n$ with $\omega_n \rightarrow 0$ not depending on the components, on which $\widehat{\mathcal{S}}(t) = \mathcal{S}$ for every t with $\bar{\Delta}_n \leq t < \nu_{\min,n} - \bar{\Delta}_n$ $\bar{\Delta}_n^{\mathcal{G}} \leq t < \nu_{\min,n} - \bar{\Delta}_n^{\mathcal{G}}$; here $\bar{\Delta}_n \leq c(\alpha, \alpha') \rho_n$ $\bar{\Delta}_n^{\mathcal{G}} \leq c(\alpha, \alpha') \rho_n^{\mathcal{G}}$ is an explicit bound on $\max_{(\ell,m)} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, given in the proof (Section S7 of the Supplementary Material).

Parts (i) and (ii) hold also at the resolution of Theorem 1, with ρ_n $\rho_n^{\mathcal{G}}$ replaced by $\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$ $\tilde{\rho}_n^{\mathcal{G}} = n^{-s/(2s+1)} (\log n)^{(2/\pi-1)+/\{2(2s+1)\}}$, the square root of the rate of that theorem, both in the condition on t_n and in Assumption 6. Since $\tilde{\rho}_n \leq \rho_n$ $\tilde{\rho}_n^{\mathcal{G}} \leq \rho_n^{\mathcal{G}}$ for large n , with a gap in the exponent when $\pi < 2$, the separation required is weaker exactly where the components are spatially inhomogeneous. Part (iii) is stated at the resolution of Theorem 3, where its bound is explicit. The componentwise rate of the lasso at its own resolution is $(\log n/n)^{s'/(2s'+1)}$ (Corollary S8.1 in the Supplementary Material), so that the separation required here is weaker than for the lasso by the factor $(\log n)^{s'/(2s'+1)}$.

Part (i) uses no separation: removing the null blocks is free, and Assumption 6 is spent entirely on not losing a small active block. Under a fixed target $\nu_{\min,n}$ is a positive constant and the assumption holds automatically, because $\rho_n \rightarrow 0$ $\rho_n^{\mathcal{G}} \rightarrow 0$; it carries content in a triangular array, where it says how fast the smallest active block may shrink. It

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bears on the components and not on $(\mathbf{X}, \mathbf{U})$, so that no design condition beyond Proposition 2 and no irrepresentability condition enters; a separation condition on the components plays the same role in the selection of variables in nonparametric additive models (Huang et al., 2010). The proof rests on a deterministic sandwich (Lemma S7.1 in the Supplementary Material): if every component is estimated within $\bar{\Delta}$ in $L_2[0, 1]$, then $\{\nu_{\ell m} > t + \bar{\Delta}\} \subseteq \widehat{\mathcal{S}}(t) \subseteq \{\nu_{\ell m} > t - \bar{\Delta}\}$, so that every $t \in [\bar{\Delta}, \nu_{\min, n} - \bar{\Delta})$ recovers $\mathcal{S}$, and that window cannot be widened by the argument. Two things are left out. The window involves σ , the design constant and s' , so the corollary does not say how to choose t_n from the data (Section 4.4 uses the one standard error rule of cross-validation, for which no theorem is given here); and it says nothing about which wavelets are active inside a block.

Thresholding is also the estimator used for prediction (Section 4.4). Write $\widehat{\boldsymbol{\theta}}^{(t)}$ for $\widehat{\boldsymbol{\theta}}$ with the blocks outside $\widehat{\mathcal{S}}(t)$ set to zero, and $\widehat{g}_{\ell m}^{(t)}$ and $\widehat{f}^{(t)} = \mathbf{A}\widehat{\mathbf{c}} + \mathbf{B}\widehat{\boldsymbol{\theta}}^{(t)}$ for the resulting components and fit, so that the levels and the blocks kept are those of the fit. Setting to zero an active block whose fitted norm is at most t costs at most its norm, which is at most $t + \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, setting to zero a null block only reduces the error, and the blocks kept do not change (Lemma S7.2 in the Supplementary

Material). With the rates of the fit this gives the risk of the thresholded estimator.

Corollary 3 (Risk of the thresholded estimator). *Let $(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}})$ be either the fit of Theorem 3, with $r_n = \rho_n^{\mathcal{G}}$, or the lasso at the resolution of Theorem S8.2 in the Supplementary Material, with $r_n = (\log n/n)^{s'/(2s'+1)}$, and let $t_n \geq 0$.*

- (i) *With no further assumption, $\sum_{\ell,m} \|\widehat{g}_{\ell m}^{(t_n)} - g_{\ell m}\|_{L_2[0,1]}^2 = O_p(r_n^2 + t_n^2)$ and $\left\| \widehat{f}^{(t_n)} - f \right\|_n^2 = O_p(r_n^2 + t_n^2)$.*
- (ii) *If Assumption 6 holds with r_n in place of $\rho_n^{\mathcal{G}}$ and $\limsup_n t_n/\nu_{\min,n} < 1$, then with probability tending to one $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t_n)$,*

$$\sum_{\ell,m} \|\widehat{g}_{\ell m}^{(t_n)} - g_{\ell m}\|_{L_2[0,1]}^2 \leq \sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$$

and

$$\left\| \widehat{f}^{(t_n)} - f \right\|_n^2 \leq \frac{4\Lambda}{\gamma} \left\| \widehat{f} - f_{J_n} \right\|_n^2 + 2 \|\mathbf{b}\|_n^2,$$

so that the thresholded estimator keeps the rate r_n^2 of the fit. If moreover $t_n/r_n \rightarrow \infty$, then $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$ with probability tending to one, and the null blocks no longer contribute to the error of the components.

Under a fixed target Assumption 6 holds automatically, and (ii) holds for every $t_n \rightarrow 0$. The factor $4\Lambda/\gamma$ in (ii) bounds the condition number of $\widehat{\Sigma}$, through which the bound passes from prediction to coefficients and back; it

is a price of the argument, and the corollary says what any threshold costs, not which threshold a data driven rule selects.

4. Computation and Tuning

4.1 The design matrix and the solver

The design (2.4) is built one block at a time: the $n \times N_J$ matrix of the wavelets of U_m is evaluated once per modulator, by the cascade algorithm of Daubechies and Lagarias (1991) [as implemented in the WaveBased package \(Montoril, 2026\)](#), and then multiplied rowwise by X_ℓ . Since a compactly supported wavelet of level j is nonzero on an interval of length $(L-1)2^{-j}$, at most $L-1$ of the 2^j columns of a level are nonzero at a given observation, so a block has at most $n(L-1)J$ nonzero entries out of nN_J and the design is stored in sparse form as soon as J is moderate. Evaluating the basis once on a grid and looking the values up removes the cascade from the inner loop of a simulation.

Problem (2.5) is then a ~~lasso~~[group lasso](#) with p unpenalized coordinates, which ~~coordinate descent~~[group descent \(Breheny and Huang, 2015\)](#) solves along a path of penalty levels; we use the ~~glmnet~~[grpreg](#) package ~~(Friedman et al., 2010)~~ [\(Breheny and Miller, 2026\)](#) of R (R Core Team, [2026](#)). Two points of scale deserve care, because they are what ties the

4.1 The design matrix and the solver

software to the theory. First, the solver standardizes the columns and orthonormalizes each chunk before penalizing it (Simon and Tibshirani, 2012), so that what it computes is (2.5) with $\|\boldsymbol{\theta}_G\|_2$ replaced by $\|Q_G^{1/2}\boldsymbol{\theta}_G\|_2$, where Q_G is the Gram matrix of the columns of the chunk after centering, which is the estimator of Remark S5.2 in the Supplementary Material; and its objective, $(2n)^{-1}\|\mathbf{Y} - \mathbf{A}\mathbf{c} - \mathbf{B}\boldsymbol{\theta}\|_2^2 + \lambda \sum_G w_G \|Q_G^{1/2}\boldsymbol{\theta}_G\|_2$, is half that of (2.5), so that the penalty level it reports is the λ of (2.5), with no conversion. Second, the solver always fits an intercept of its own, which carries the level of the constant covariate when there is one. For the lasso option, solved by coordinate descent with the `glmnet` package (Friedman et al., 2010), the penalty level reported for a fit is the λ of the lasso problem and not the one of the solver, which rescales its own by the ratio between the total number of columns and the number of penalized ones when some coordinates carry zero penalty; the conversion is checked by verifying the Karush-Kuhn-Tucker conditions of (2.5) at every point of the path. Second, and when one linear covariate is the constant, its column has zero variance, and the solver may either drop it or share the level with its own intercept; the fit is the same either way, but the level c_ℓ of the constant covariate is only recovered after the intercept is added back to it. For both penalties the scale is checked by verifying the Karush-Kuhn-Tucker conditions of the

problem at every point of the path.

4.2 Rescaling and the boundary

The theory places $\mathbf{U}$ in $[0, 1]^q$ by assumption. In practice each modulator is mapped to that interval by a monotone transformation, and. By default the image is $[0, 1]$ itself, which is the setting of Section 3. Optionally, as in the wavelet estimators of Montoril, Chang, and Vidakovic (2019), the image is $[\epsilon, 1 - \epsilon]$ with ϵ of the order 2^{-J} fixed, which keeps the observations away from the strip point where the periodization of (2.3) leaves an artifact joins the two ends of the interval; the price is that the fitted component integrates to zero over the rescaled range and not over $[0, 1]$, a shift of level and not of shape. The margin is a finite sample device, and the reason to allow it is approximation: with the data away from the ends, a component needs to be regular only on its support, where it can be replaced by a periodic extension that matches at $0 \equiv 1$, built from a bounded extension operator (Triebel, 1983, Chapter 3). This removes the stall of Proposition S3.1 and restores the bias of Lemma 2, with a Besov constant of order $\epsilon^{-(s-1/\pi)}$, an order that, by the embedding of Besov into Hölder spaces, cannot be improved when the component takes different values at the two ends of its support. Two features keep the margin out of the theory. It has to be fixed, since with

4.3 Choosing the resolution and the penalty

ϵ of the order 2^{-J} the exponent guaranteed for the bias is no better than that of periodization without a margin; and it degrades the design, since the wavelets supported inside the margin are invisible to the data, and the smallest eigenvalue of the Gram matrix of the basis restricted to the support of the data vanishes as soon as $2^J \geq (L - 1)/(2\epsilon)$, so that Proposition 2 does not hold as stated. The alternative is the basis adapted to the interval of Cohen, Daubechies, and Vial (1993), which reproduces polynomials on $[0, 1]$ and characterizes Besov regularity without any periodicity (Cohen, 2003, Section 3.9), so that Lemma 2 holds with the Besov space of the interval and the same proof. Its cost is structural: the coarsest level must be large enough for the two boundary zones to be disjoint, the constant is no longer a single column but a combination of the scaling functions, and the constraint of (2.2) has to be imposed by reparametrization, which leaves $pq(2^{j_0} - 1)$ unpenalized parameters instead of none. We therefore keep the periodized basis as the default and offer the interval basis as an option.

4.3 Choosing the resolution and the penalty

The pair (J, λ) is chosen jointly. The default is K fold cross validation on a grid of resolutions, with the folds drawn once and reused for every J , one design and one penalty path per resolution, and the usual two readings of

4.3 Choosing the resolution and the penalty

the curve: the minimizer of the cross validated error and the largest penalty within one standard error of it; the threshold of Section 4.4 is applied to the fit at the minimizer. In the software the whole procedure, from the cross validation over (J, λ) to the threshold, is the single call `cv.wafc(x, u, y)`. Two cheaper rules are also available for the lasso option, the Bayesian information criterion and its extended version, with degrees of freedom equal to the number of nonzero coefficients plus the p levels, which is the usual unbiased count for the lasso.

The theory supplies a fourth rule, since (3.4) and (3.1) are explicit. It is not a practical competitor as it stands, because it needs σ , which has to be estimated, and s' , which is unknown, and because it is a statement about order, so that at moderate sample sizes it can sit above the resolution that minimizes the realized error, the pre-asymptotic regime being wide when s' is small. In our experiments with the lasso option the penalty (3.1) of its theory is 7 to 13 times the one that minimizes the realized error, and it is the penalty, more than the resolution, that makes the rule costly; the resolution (3.4) of its theory misses the best one by up to two levels, above or below according to s' . Its interest is as a reference for what the cross validated choice ought to approach, and the difference between the two is a finite sample quantity worth measuring. The four rules are compared in

the simulation study.

A remark on the ~~structural variant~~ other penalties. The lasso option, with $\|\boldsymbol{\theta}\|_1$ in place of $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$, is fit by `glmnet`; its theory is in Section S8 of the Supplementary Material, where its rates carry an extra logarithmic factor, and the threshold of Section 4.4 applies to it unchanged. Replacing $\|\boldsymbol{\theta}\|_1$ in (2.5) by the sparse group lasso penalty of Simon et al. (2013), with one group per block (ℓ, m) and fit with the `sparsegl` package (Liang et al., 2024), adds a second tuning parameter and removes entire components rather than isolated ~~coefficients~~ chunks, which is what a reader interested in knowing which modulators act on which coefficients would want, although for the lasso itself (2.5) that question is already answered by thresholding the block norms (Corollary 2). It is a different estimator, with a different oracle inequality (Lounici et al., 2011), and the theory above does not cover it; the choice between the two is settled empirically.

4.4 Choosing the threshold

The threshold is chosen on the folds of Section 4.3, at the pair (J, λ) already selected: each fold is refitted at that pair, with no new search, thresholded at every candidate t and scored on the observations it left out. A thresholded fit is constant in t between two consecutive block norms, so the candidates

are exhaustive and not a grid. The default is the largest t whose cross validated error is within one standard error, between folds, of the smallest, the one standard error rule of the penalty carried to the threshold; the minimizer of the cross validated error is available as an option aimed at prediction. The fit is not refitted after the threshold: the levels and the blocks kept are those of the fit, which is the estimator of Corollaries 2 and 3. Neither rule has a theorem here: Corollary 2 gives the window of thresholds that recovers the structure, and Corollary 3 what any threshold costs.

Supplementary Materials

The online Supplementary Material contains the proofs of all the results stated in the paper, together with the auxiliary lemmas they use: the identification argument and the properties of the periodized basis, the approximation lemmas in L_2 and in the supremum norm and the proposition on the cost of periodization, the population and empirical bounds for the Gram matrix of the product design, the ~~four lemmas~~ properties of the partition into chunks, the calibration of the penalty over chunks and the other lemmas behind the oracle inequality and its form for the estimator that the software computes, and the ~~two lemmas~~ lemmas behind the rates, and the

~~lemma~~ two lemmas behind the recovery of the block structure by thresholding and the risk of the thresholded estimator, and the theory of the coordinatewise lasso, with its proofs.

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Write the acknowledgements here.

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